Radar Overview

Prof. Dr. Ali Kara

Objective

learning fundamental principles of radar operation

Learning Outcomes

At the end of the lesson, you should be able to

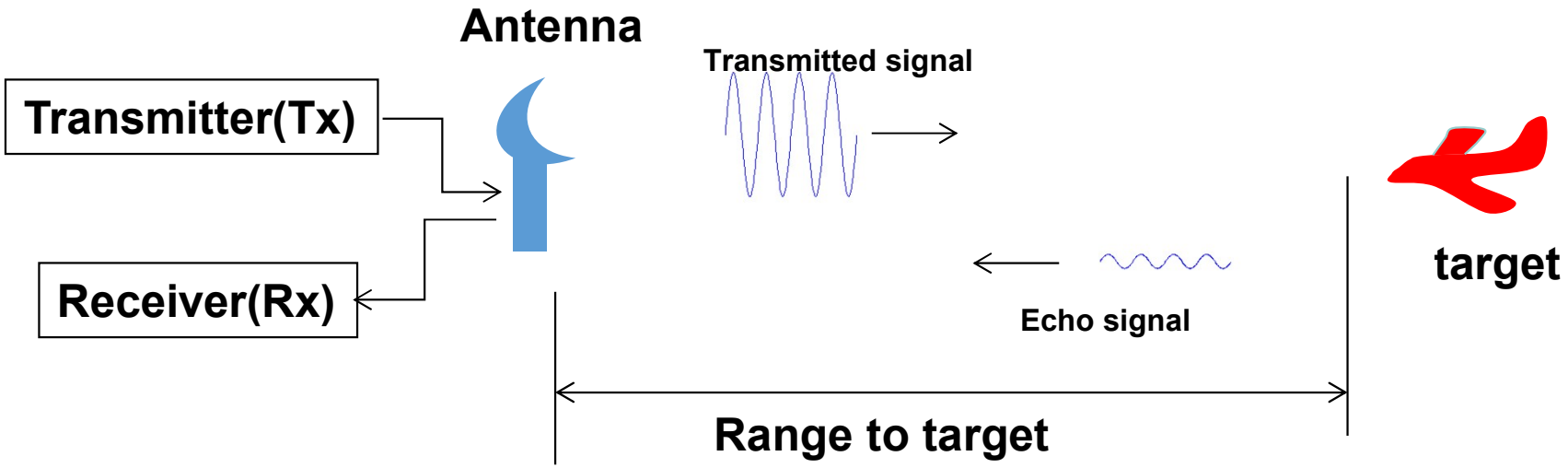
1. describe operation of a radar system
2. describe typical radar signal in time and spectral domain
3. list some major design parameters of radars
4. derive simple radar range equation
5. describe radar antenna and scanning types
6. describe major radar parameters required in EW operations

Presentation Plan

- Radar Principle
- Radar Signaling
- Radar System Parameters
- Radar Detection
- Operational Aspects

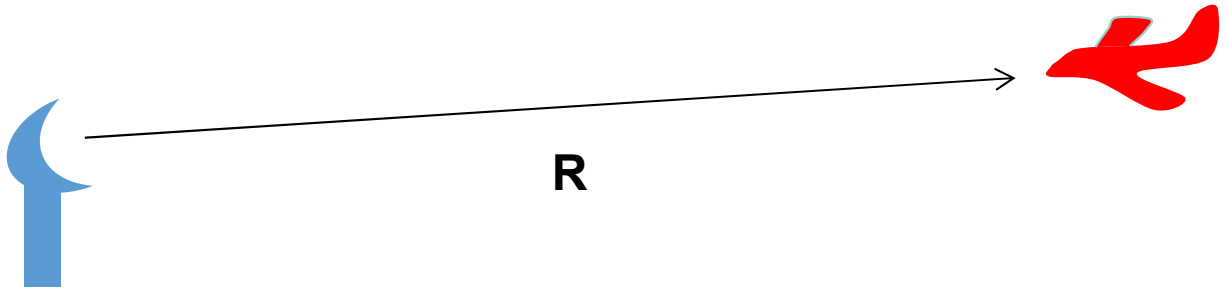
Radar principle

- Radar: Radio Detection And Ranging



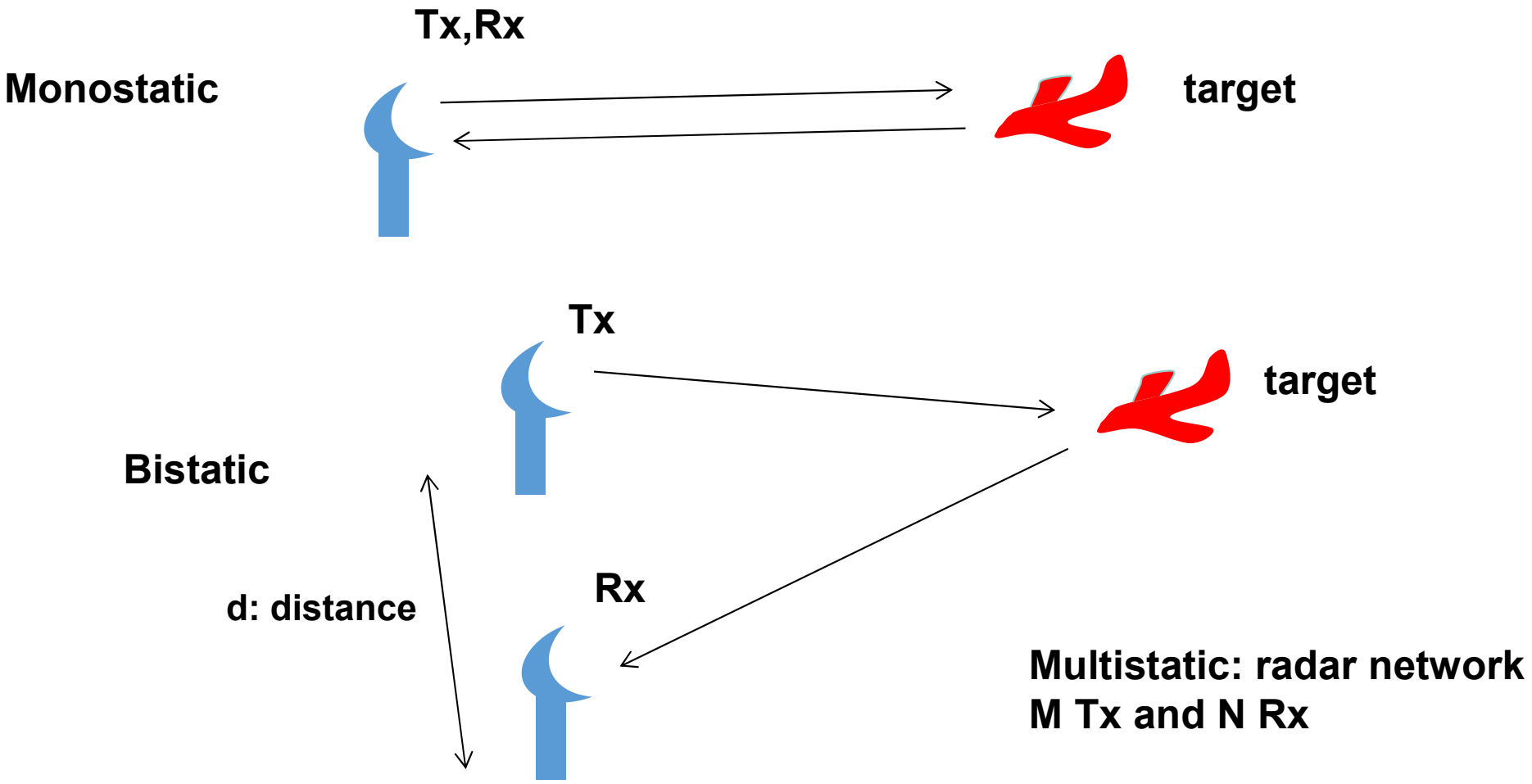
Radar principle

Range to a target

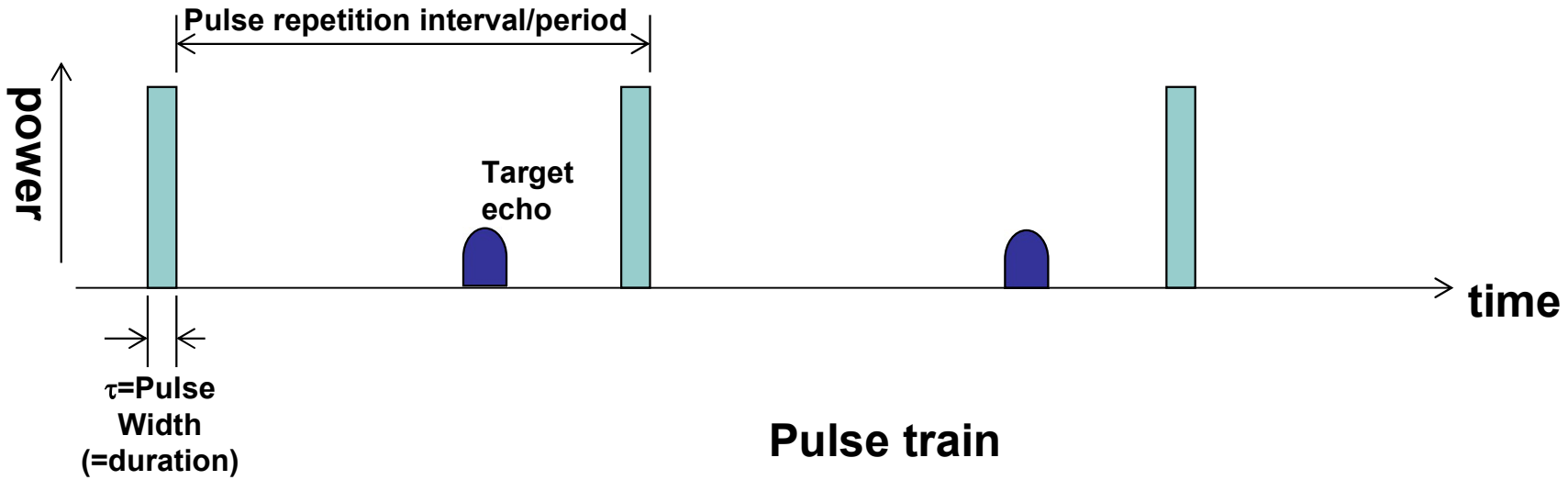


- T_R: time the radar signal to travel to target and back
- c=the speed of light = the speed of electromagnetic waves
- R=range to a target
  - $R=cT_R/2$

Monostatic and bistatic Radar



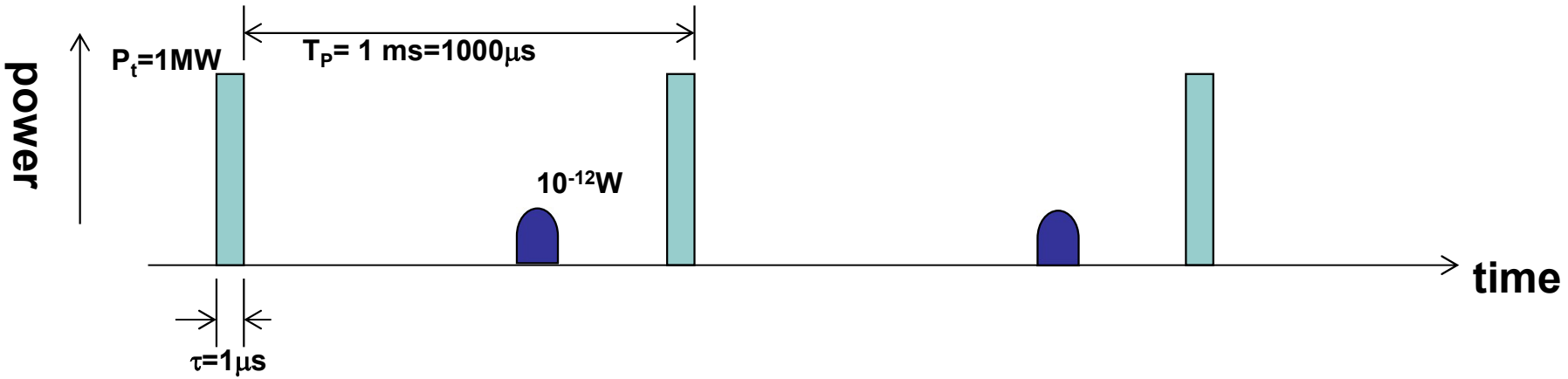
Radar pulse train (envelope)



Pulse train

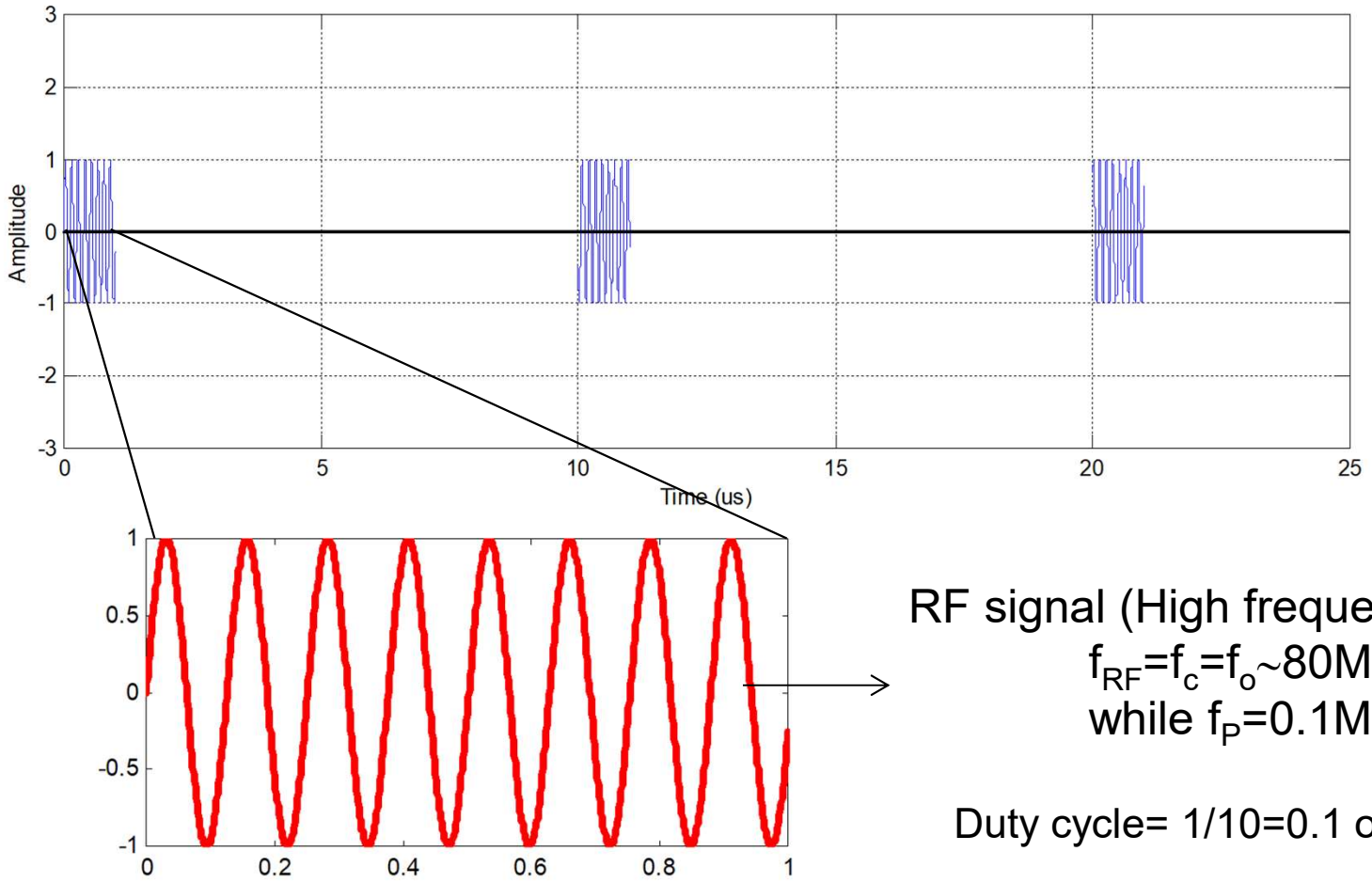
- $T_P = 1/f_P$ =Pulse repetition interval (PRI)
- f_P=pulse repetition frequency(PRF)
- P_{av}=average power=$P_t \tau/T_P = P_t \tau f_P$
- τ/T_P =Duty cycle

Medium range air-surveillance radar



- $f_P=1/T_P$=PRF=1/PRI=1000Hz=1kHz
- Duty cycle=1/1000=0.1%
- Power attenuation=10^{15}=150dB

Radar waveform: temporal



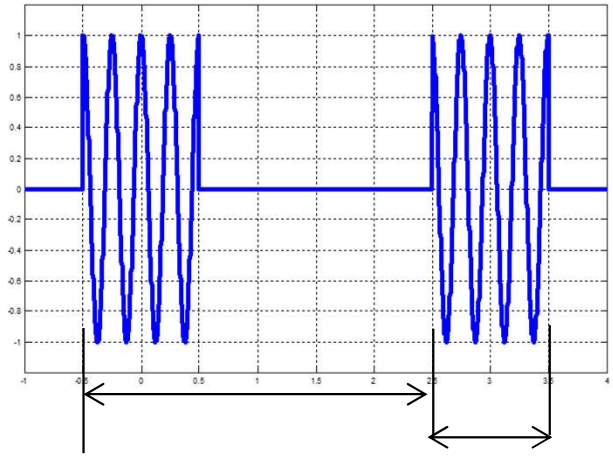
RF signal (High frequency carrier)

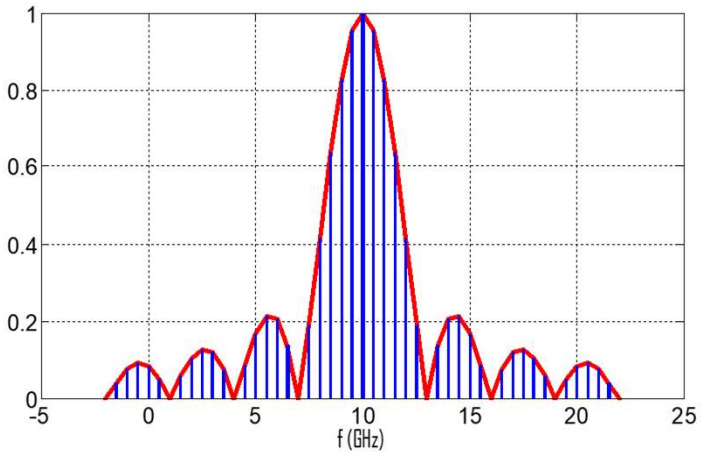
$f_{RF}=f_c=f_o$~80MHz

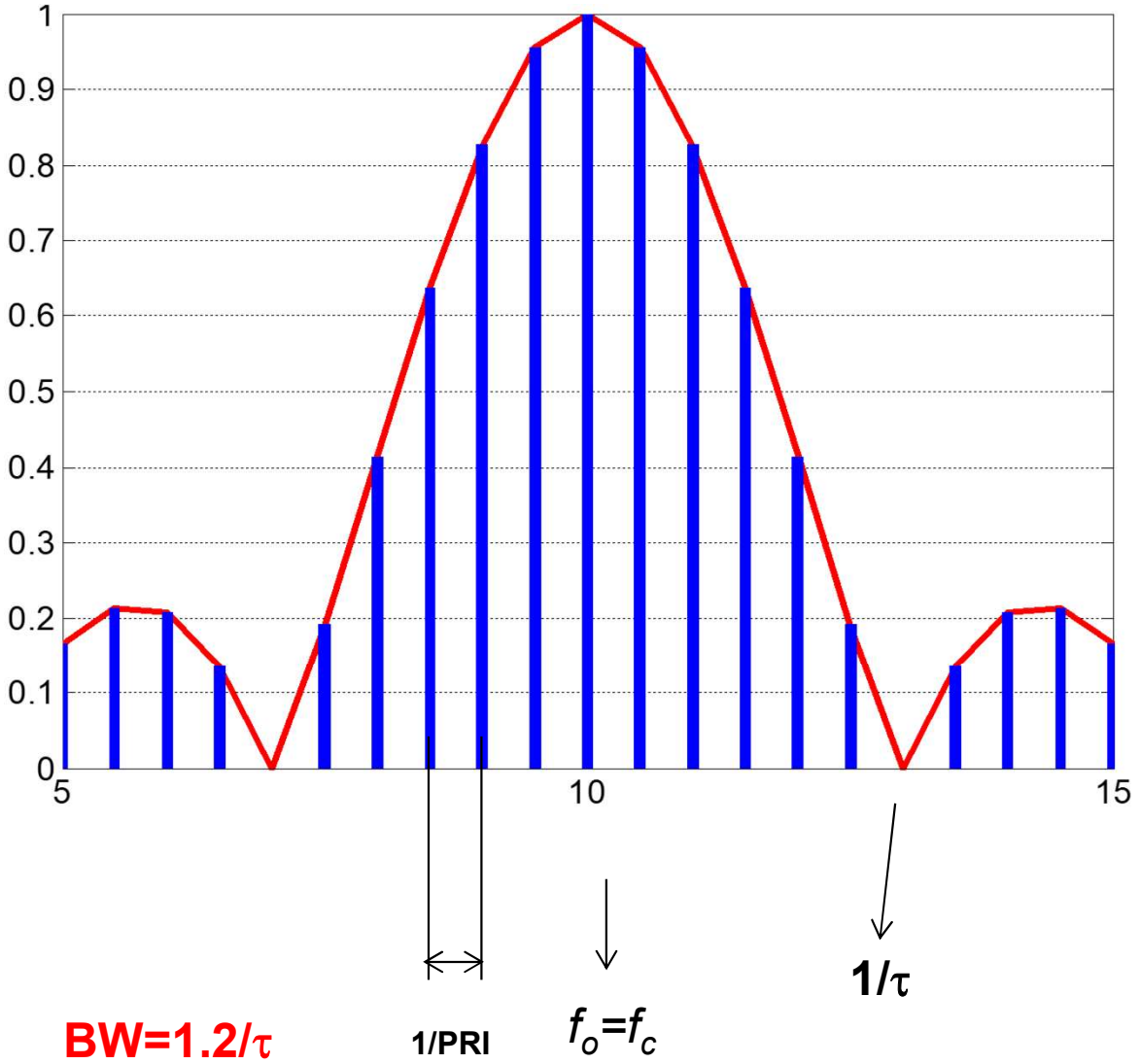
while f_P=0.1MHz

Duty cycle= 1/10=0.1 or 10%

Radar waveform: temporal & spectral







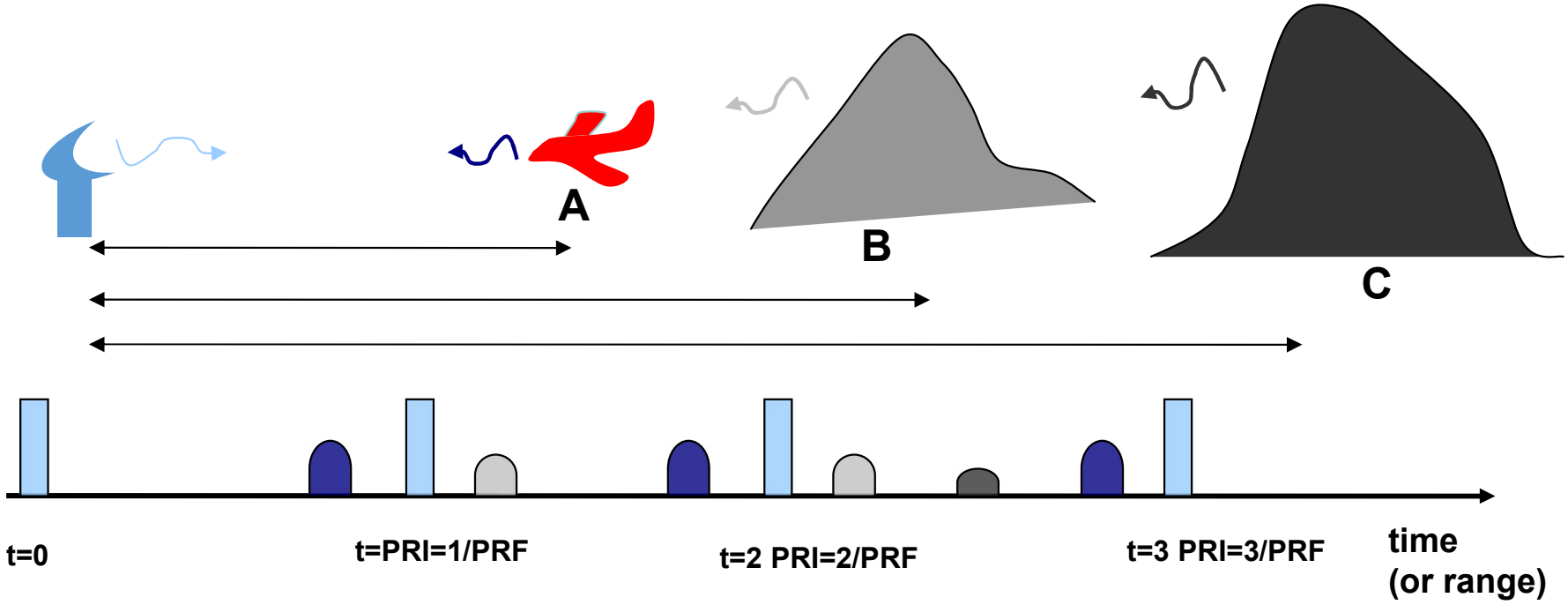
Maximum Unambiguous Range

- Transmit a pulse, and listen for an echo from target
- After receiving the echo pulse, transmit another pulse, and listen for another echo and so on
- Determine the "time difference" between the consecutive echo pulses (pulse repetition interval-PRI or PRF)
- Time between the pulses is too short, then an echo signal from a long range target might arrive after the transmission of the next pulse, and this introduces an incorrect or "ambiguous range" measurement
- Echoes arriving after transmission of the next pulse is called second time around echoes (multiple around echoes)
- The maximum unambiguous range is the range beyond which targets appear as second time around echoes, and is given by

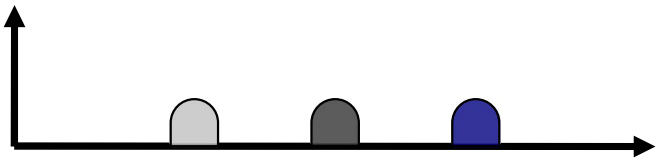
$$R_{un}=cT_P/2=c/2f_P$$

Range ambiguity

$$f_p = \frac{1}{PRI} = \frac{c}{2R_{un}}$$



Display



Resolving range ambiguity: multiple PRF

PRF 1: $f_1 \rightarrow R_{un1}$

target apparent range measured by $f_1 : R_1$

True range:

$$R_{true} = R_1, R_1 + R_{un1}, \text{ or } R_1 + 2R_{un1} \text{ or } R_1 + 3R_{un1}, \text{ or } ,..$$

PRF 2: $f_2 \rightarrow R_{un2}$

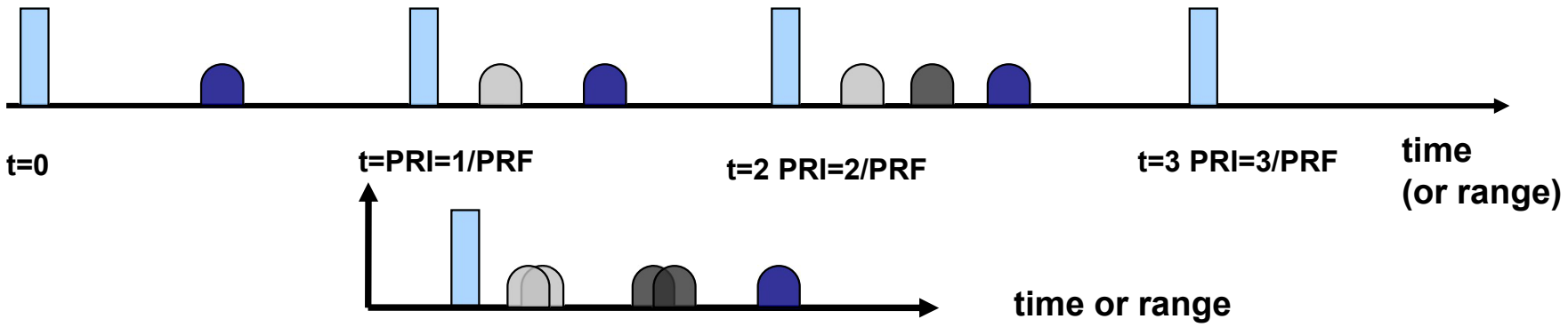
target apparent range measured by $f_2 : R_2$

True range:

$$R_{true} = R_2, R_2 + R_{un2}, \text{ or } R_2 + 2R_{un2} \text{ or } R_2 + 3R_{un2} \text{ or } ,..$$

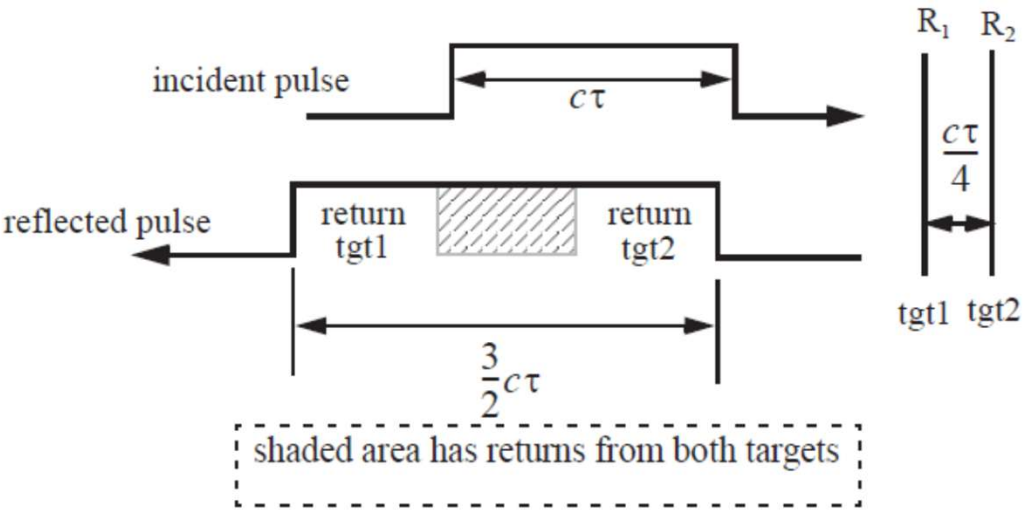
Usually, three PRF is used to reduce possible errors in range ambiguity(two is enough in theory)

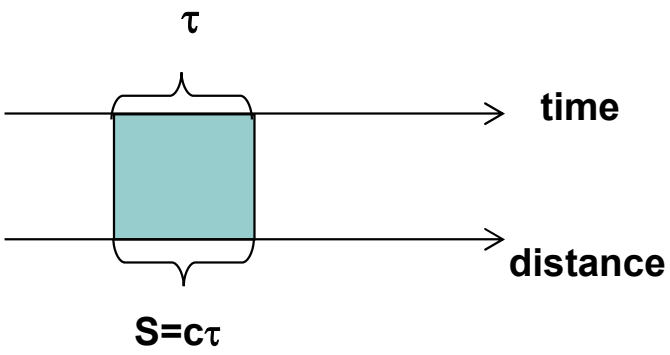
The correct range is that value which is the same with the two PRFs !

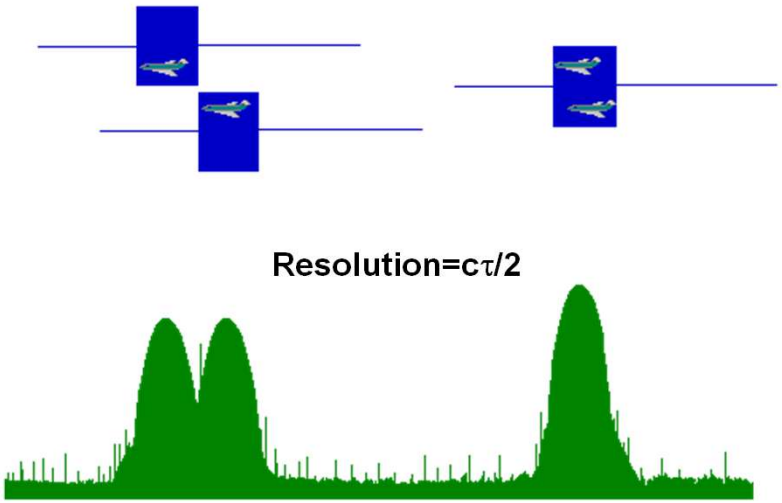


Range resolution

- The distance over which a waveform extends in space is $c\tau$
- Two equal size targets can be recognised as being resolved in range when they are separated by a distance of $c\tau/2$
- (½ is introduced due to two way travel of the signal)







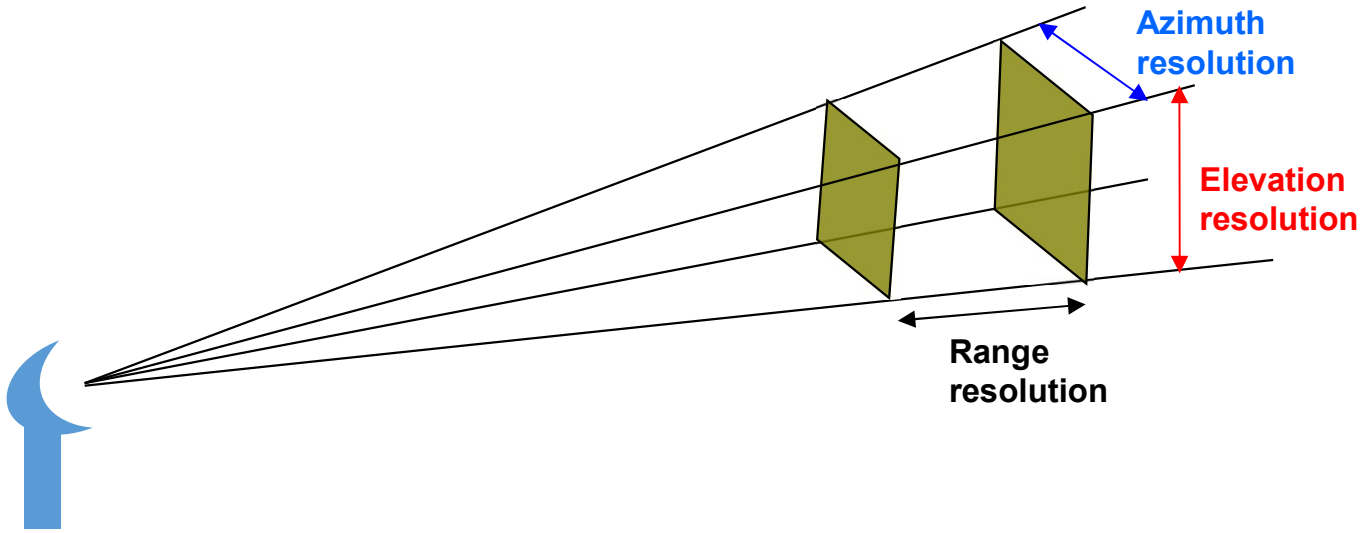
τ=1μs, two equal targets can be resolved if they are separated by 150m

Example

- A typical Early Warning Radar
- PW = 5 μsec
- Range resolution=5 $.10^{-6}$ (sec) x 3. 108 (m/sec) /2=750m

- A typical Target Tracking (Pulse Compression Radar)
- PW =0.1 μsec $\rightarrow$
- Range resolution=5 $.10^{-5}$ (sec) x 3. 108 (m/sec) /2=15 m

Resolution cell (3D resolution or radar cell)



Range resolution: pulse width

Azimuth resolution: antenna beamwidth in azimuth

Elevation resolution: antenna beamwidth in elevation

Example

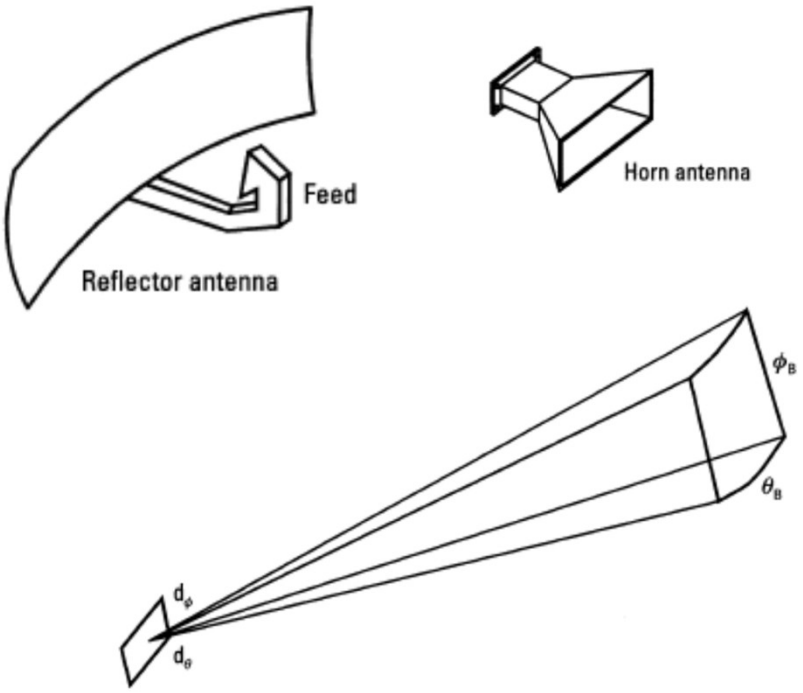
Calculate range resolutions for a target at a range of 100 km for an Early Warning Radar system

- f = 1 GHz
- PRF= 300 Hz
- PW = 4 μsec
- Antenna Beamwidth (azimuth) = 2 (deg) = 0.035 (rad)
- Antenna Beamwidth (elevation) = 5 (deg) = 0.087 (rad)

- **Range resolution = 4 (μsec) 3.0 108 (m) / 2 = 600 m**
- **Azimuth resolution = 100 000 (m) 35 10^{-3} = 3500 m**
- **Elevation resolution = 100 000 (m) 87 10^{-3} = 8700 m**

Radar Antenna: Gain and effective area

- All radars use directive antennas with narrow beamwidths
- Mechanical steering or electronical steering is used in radar systems
- Revisit time≡scan time: the time that an antenna takes to return to view the same region of space



Gain *and effective area are related by*

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi\rho_a A}{\lambda^2} \approx \frac{26000}{\theta_B \varphi_B}$$

ρ_a: *antenna aperture efficiency*

A: *antenna physical area*

φ_B, θ_B: azimuth and elevation half−power beamwidths ($HPBW$)

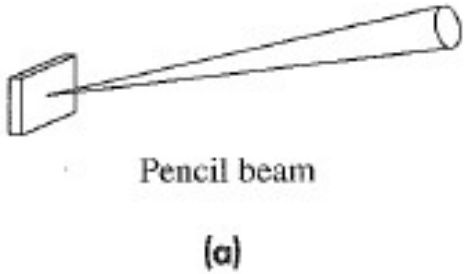
Consider *a pencil beam antenna:* $\theta_B = \varphi_B = 1^0$, *then* $G = 26000$ (*or* $44\ dB$)

Antenna parameters: beams

- $BW \approx 1^0$

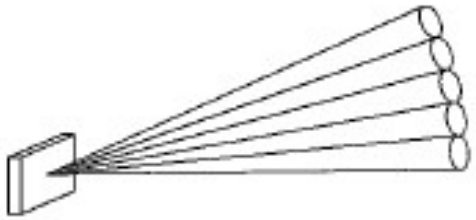
-accurate location measurements

-tracking (3-D radar) (az,el)



Pencil beam

(a)



Stacked beam

(c)

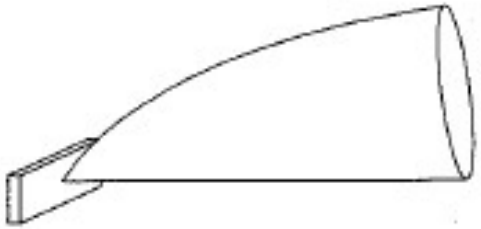
- scanning of large volume
- use 3-9 pencil beams
- stacked beams

- $BW_{az} \approx$ a few degrees
- $BW_{el} \approx 4 - 10\,x\,BW_{az}$
- 2-D radars



Fan beam

(b)



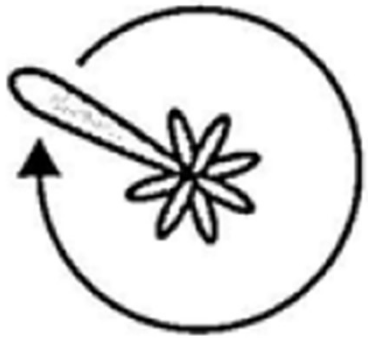
Shaped beam

(d)

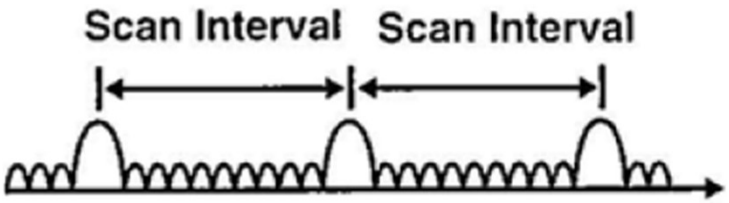
- modification of fan beam
- shaped beam: cosecant squared shaped beam

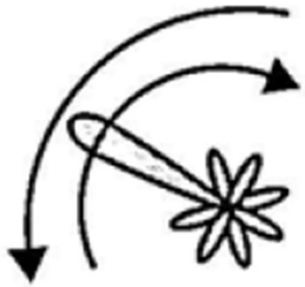
Scan types: circular and sector [Adamy]

- Provides azimuth bearing and range but not elevation
- Wide elevation beam is necessary

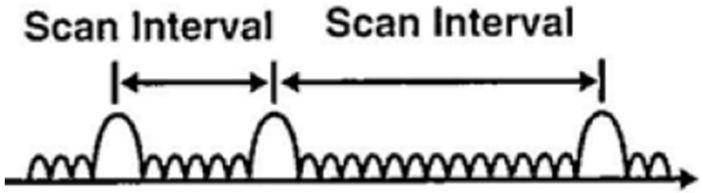


circular scan

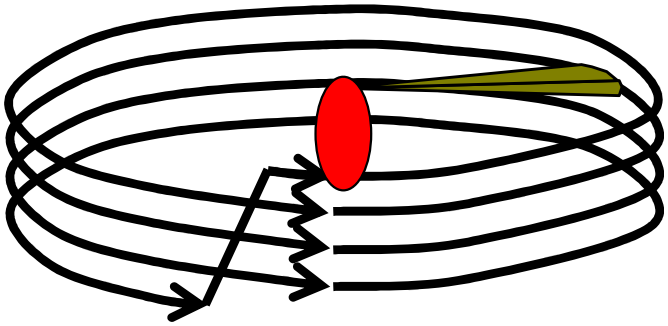




Sector scan

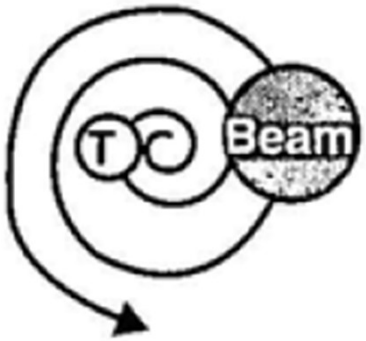


Scan types: raster, helical and spiral [Adamy]

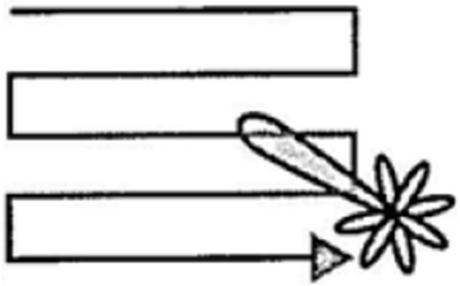


Helical scan

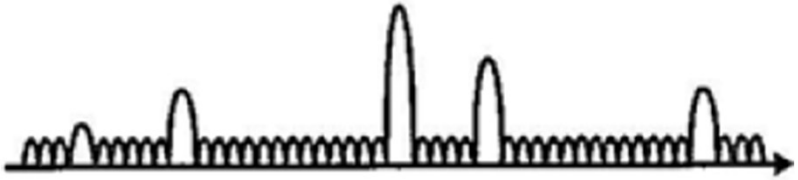




Spiral scan [Wiley]



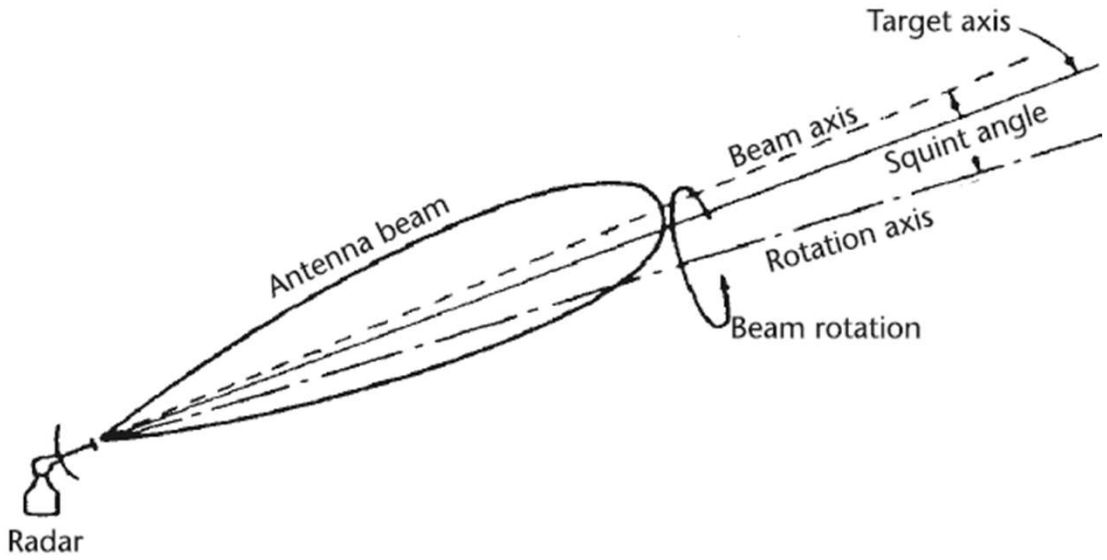
Raster scan



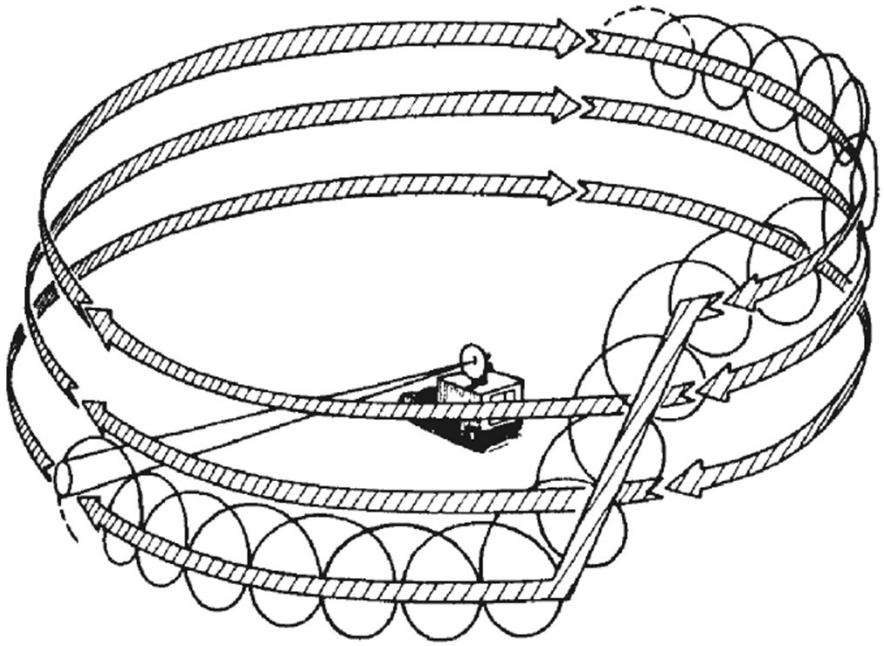


Scanning types: tracking-CONSCAN

Tracking a target whose approximate location has already been known



Conical (CONSCAN) scan [Wiley]



Helical + conical scan [Wiley]

Scanning: example

Take a X-band AI (a 1 m antenna size Air Interceptor-AI)

Beamwidth ~ $1.2\lambda/D$ ->0.035 rad. (~2 degree)

For 90 degree scan in azimuth,

45 beam positions

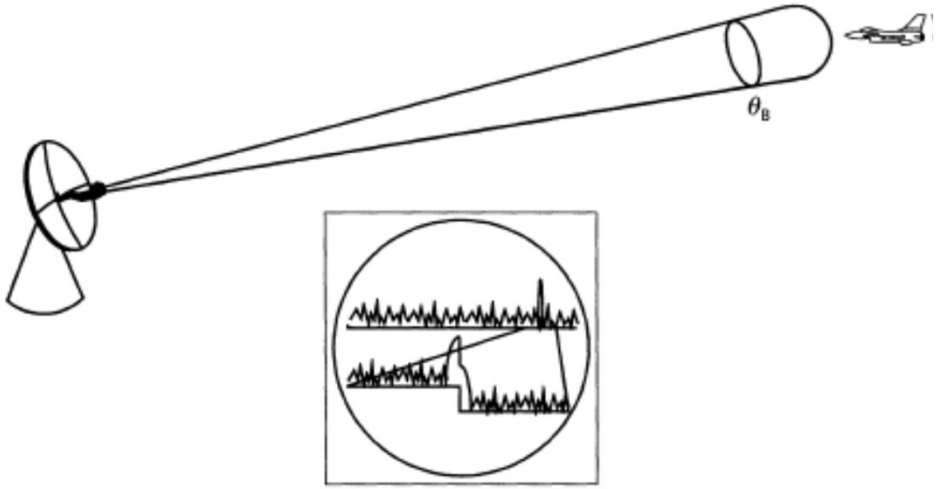
For 3 scan lines in elevation

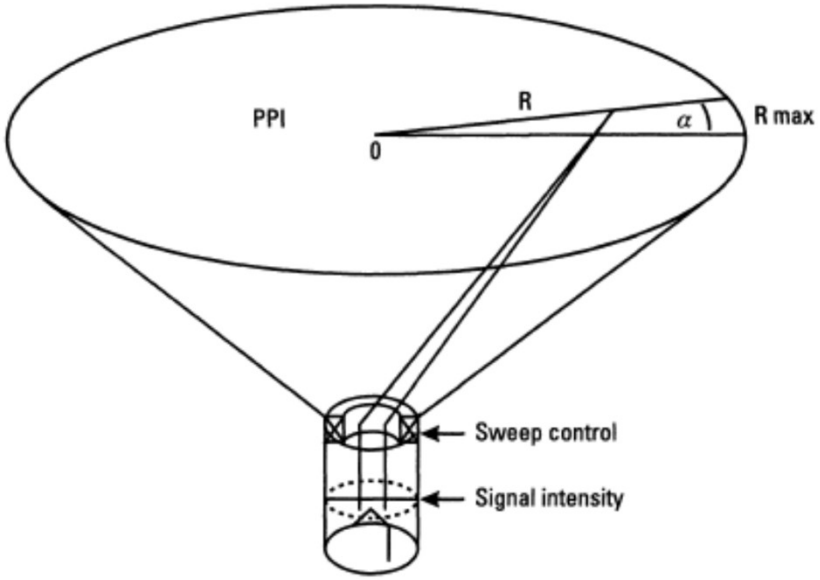
135 beam positions

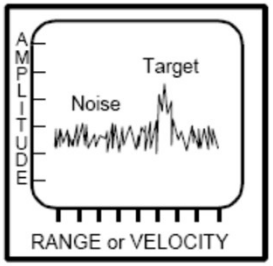
Total time for search pattern:

(dwell time on each beam position) *(number of positions)

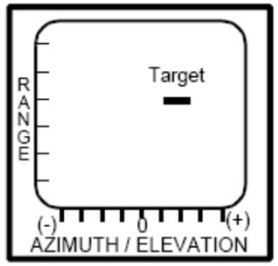
Radar Displays [Neri]







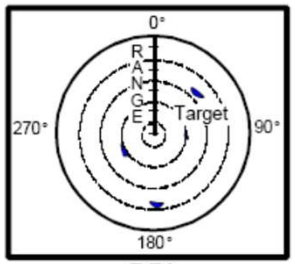
A-SCOPE



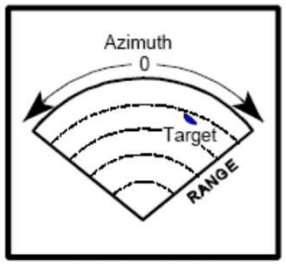
B-SCOPE / E-SCOPE



C-SCOPE



PPI

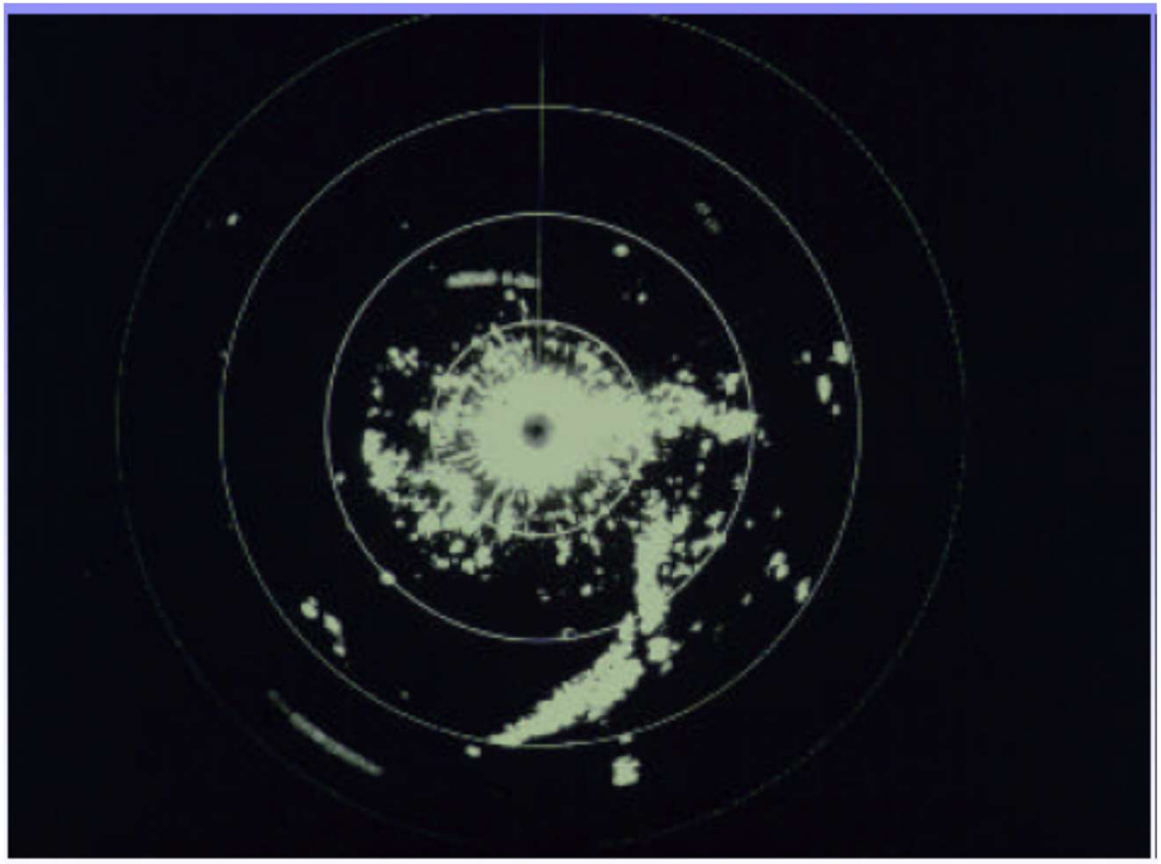


SECTOR PPI

Tracking radar
- Pencil beam
- Amplitude/range display (A/R)

Air-surveillance radar
- **Plan-position indicator(PPI)**
- Location of target in azimuth and range

Example: Radar Screen (clutter?) [Skolnik]



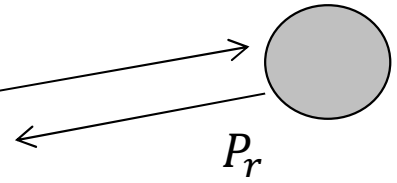
Radar cross section

- RCS of a target represents the magnitude of echo signal returned to the radar.

$$\sigma \equiv \frac{\textit{reflected power from the target}}{\textit{incident power density}} = \frac{P_r}{P_d}$$

$$= 4\pi R^2 \frac{E_r^2}{E_i^2} (m^2)$$

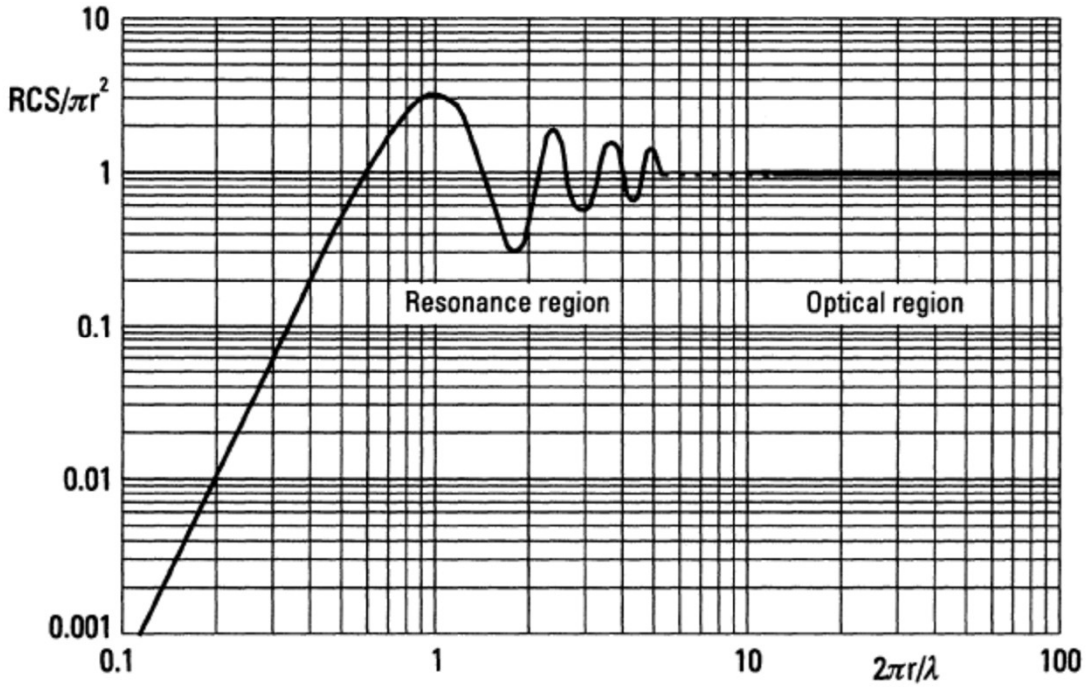
$$P \propto |E|^2$$



P_r

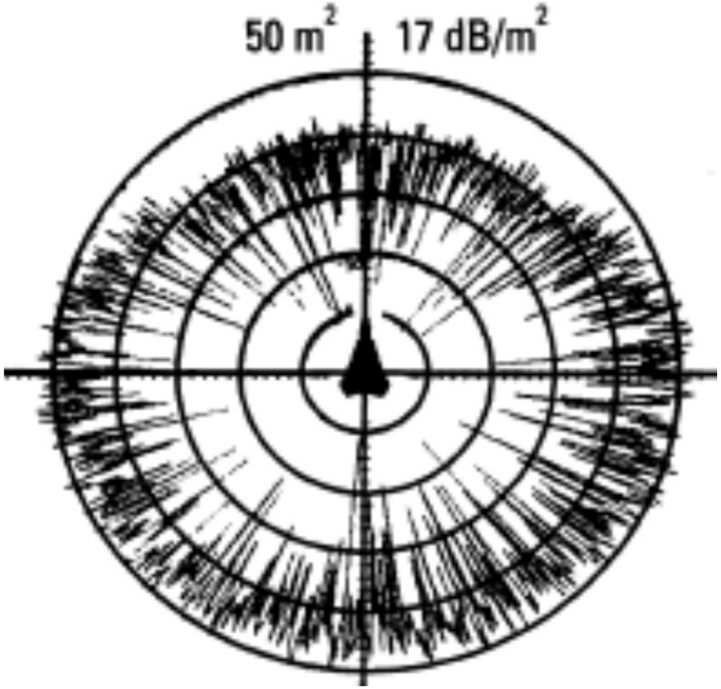
$$P_d \propto \frac{E_i^2}{4\pi R^2}$$

RCS value depends on
- Target aspect angle
- Movement
- Geometry
- polarization
- Frequency (wavelength)

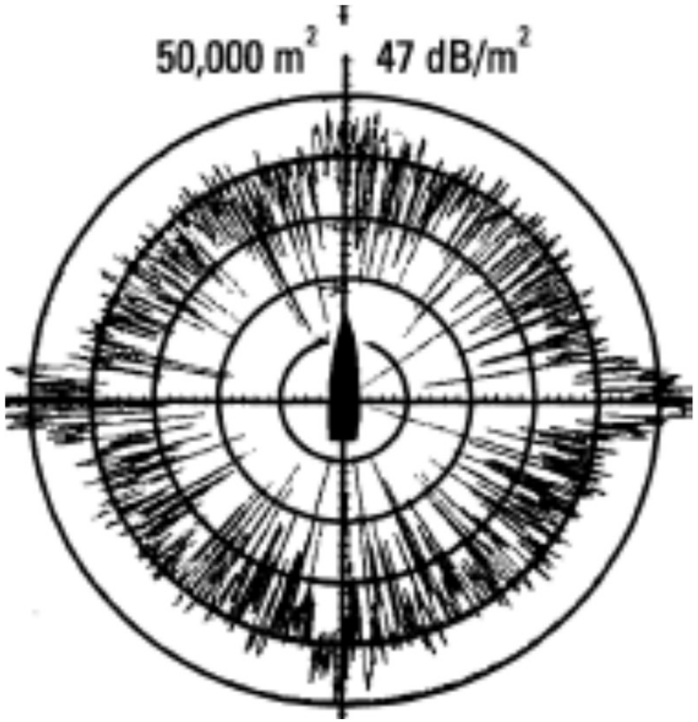


RCS of a sphere [Skolnik]

RCS of ship and aircraft [Neri]



aircraft



ship

Fighter: 0.5-5 m^2 (scan to scan fluctuations)
Missile: 0.001-0.1 m^2 (no fluctuations)

RCS Fluctuations !

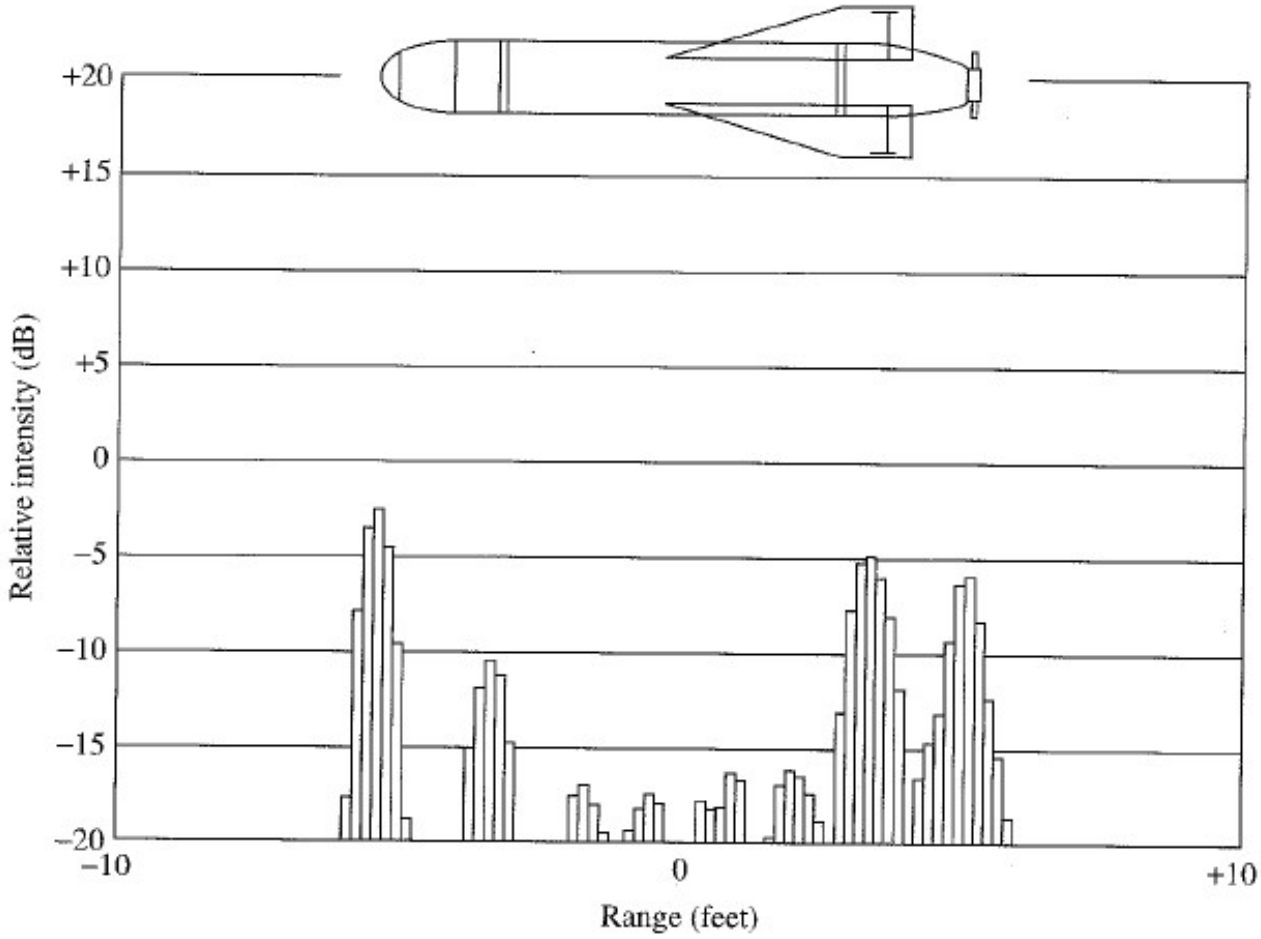
Ships: 10^2 - 10^6 m^2

$$\sigma \approx 52\sqrt{f_{MHz}}\sqrt{D^3}$$

f_{MHz}: *RF frequency*

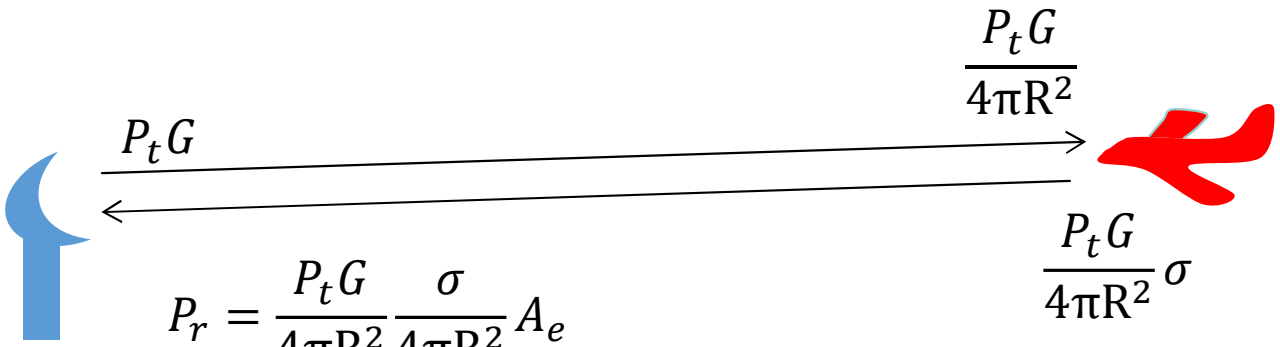
D: *displacement in kilotons*

Example: Missile RCS



High resolution "range profiles" of a small missile (individual scattering centers)

The radar range equation



$Radar\ cross\ section\ (RCS)\text{:}\ \sigma\ [dBs\ or\ m^2]$

Received *power*

$$P_r = \frac{P_t G A_e \sigma}{(4\pi)^2 R^4}$$

When $S_{min} = P_r$

$$G = \frac{4\pi A_e}{\lambda^2}$$

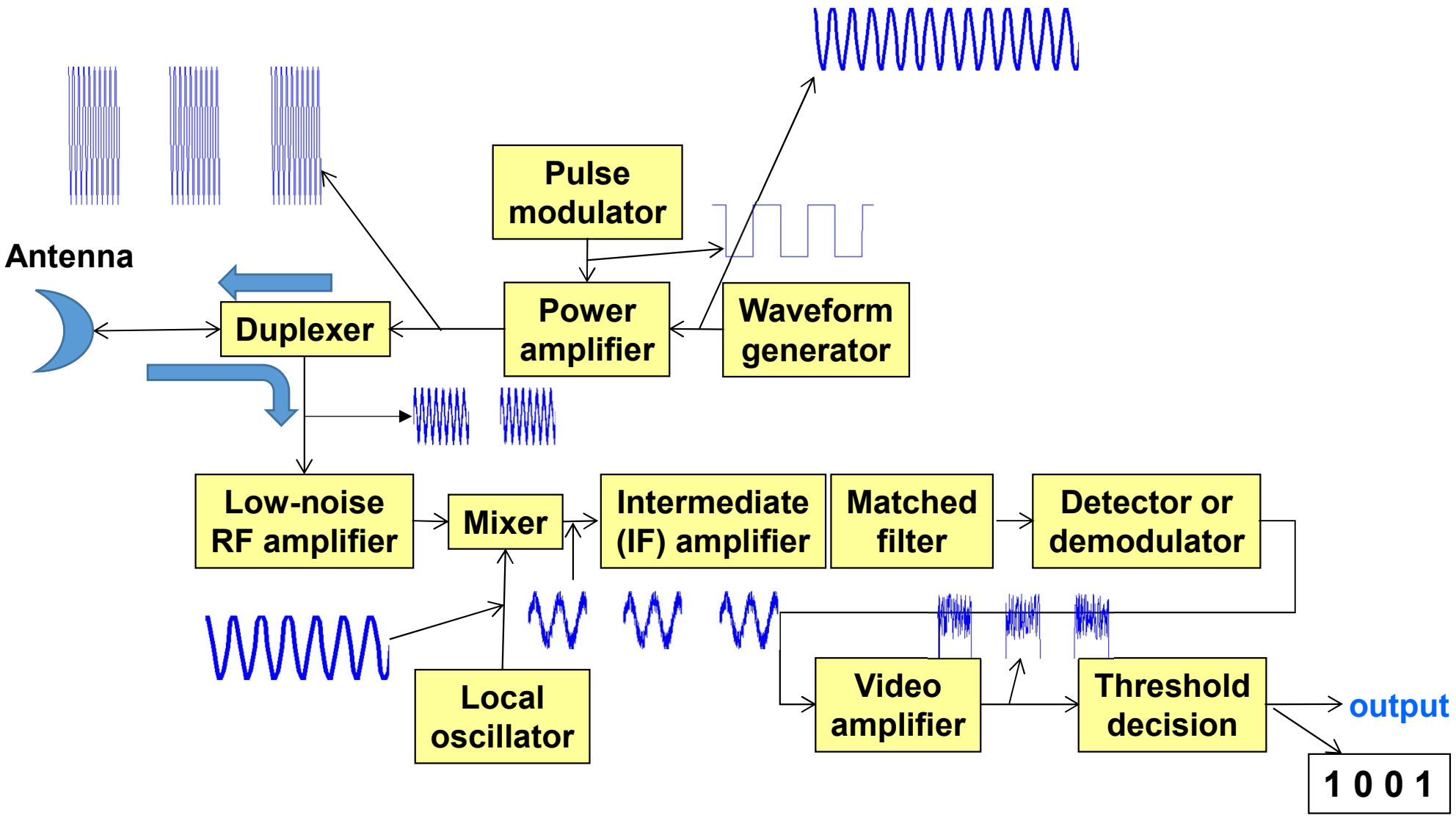
$$R_{max} = \left[\frac{P_t G \lambda^2 \sigma}{(4\pi)^3 S_{min}}\right]^{1/4} = \left[\frac{P_t A_e^2 \sigma}{4\pi \lambda^2 S_{min}}\right]^{1/4}$$

Example

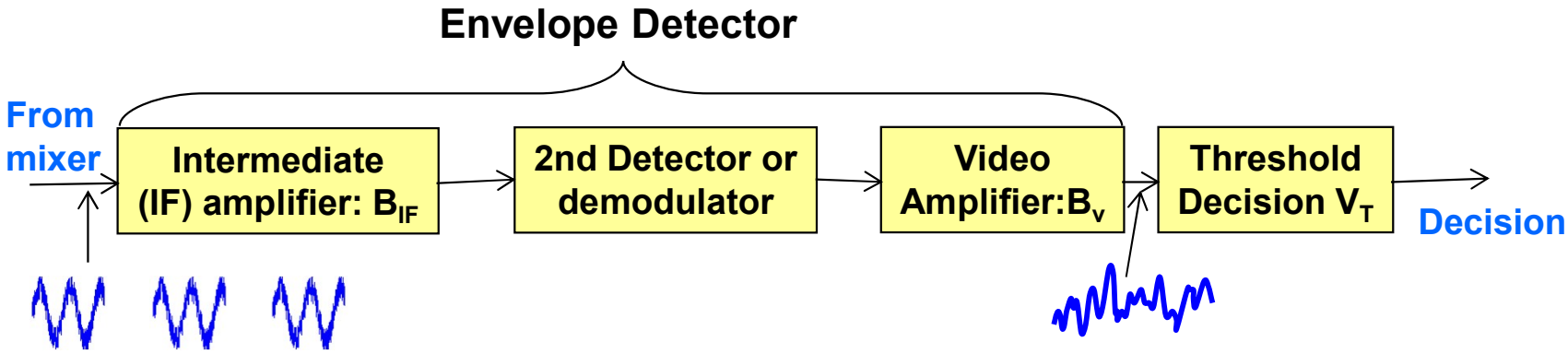
Consider a radar system

a. The maximum unambiguous range is desired to be 100 km, what is the PRF?
b. How long does it take for the radar signal to travel to the target and back when the target is at maximum range?
c. If the pulse width is 1.5us, what is the extent of the pulse energy in space?
d. How far apart in range must two equal size targets be separated (range resolution) in order to be certain they are completely resolved by a pulse width of 1.5us?
e. If the radar has a peak power of 0.8MW, what is its average power?
f. What is the duty cycle of the radar?

Pulse Radar system

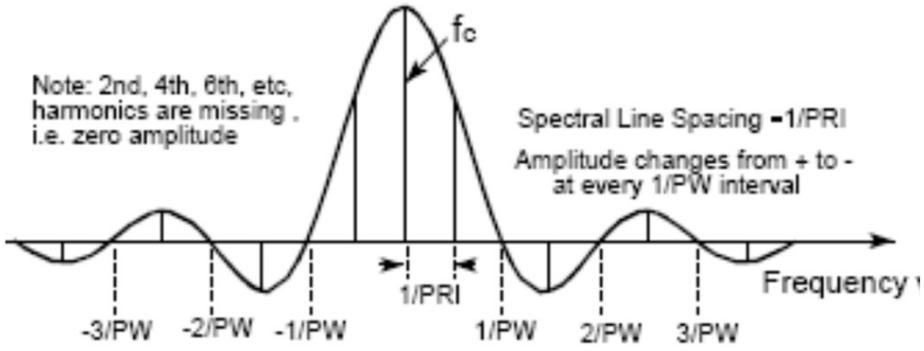


Detection section of a radar: BW



a superheterodyne (superhet) radar Rx

- $B_v \geq B_{IF}/2$
- $f_{IF} >> B_{IF}$
- **BW of the radar ≡ BW of the IF amplifier**



Radar Frequencies (RF carrier ranges)

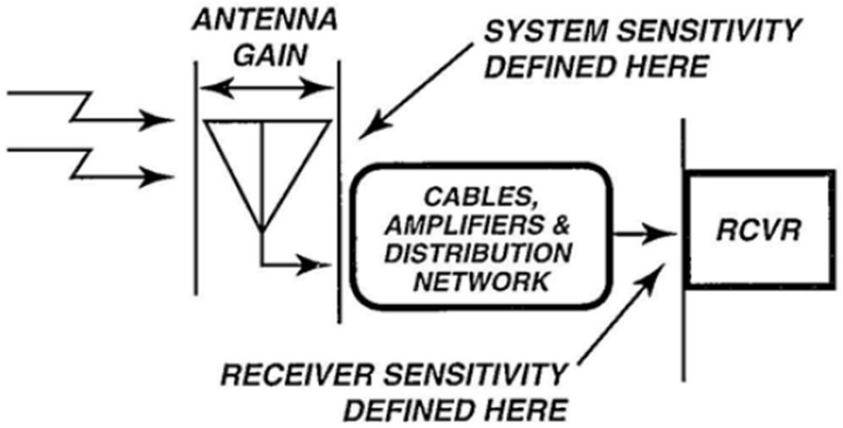
Band designation	Nominal Frequency range
HF	3-30MHz
VHF	30-300MHz
UHF	300-1000MHz
L	1-2GHz
S	2-4GHz
C	4-8GHz
X	8-12GHz
Ku	12-18GHz
K	18-27GHz
Ka	27-40GHz
V	40-75GHz
W	75-110 GHz

New Band designation

A band 0 to 0.25 GHz
B band 0.25 to 0.5 GHz
C band 0.5 to 1.0 GHz
D band 1 to 2 GHz
E band 2 to 3 GHz
F band 3 to 4 GHz
G band 4 to 6 GHz
H band 6 to 8 GHz
I band 8 to 10 GHz
J band 10 to 20 GHz
K band 20 to 40 GHz
L band 40 to 60 GHz
M band 60 to 100 GHz

Early Warning Radar: UHF, S (C-F)
Radar altimetre: S, C (E-H)
Meteorology radar: C and X
Airborne (fighter/attack): X and Ku (I,J)

Radar sensitivity↔ Noise

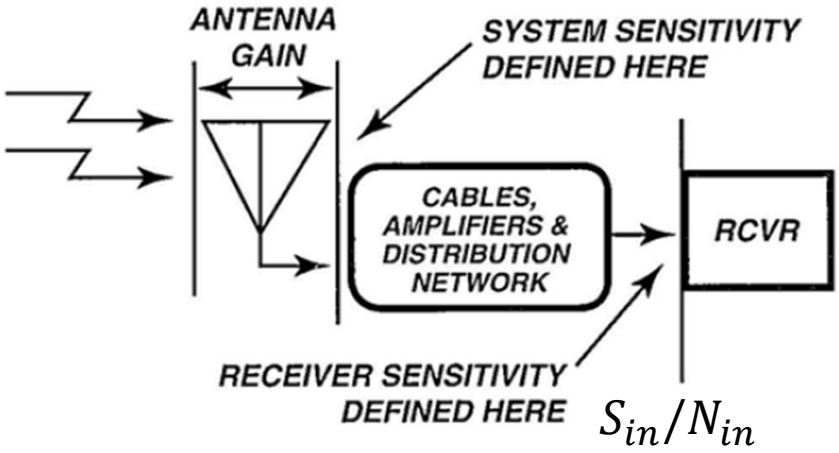


[Adamy]

- Important limiting factor of radar range: Receiver(Rx) noise
- Thermal Noise Power generated at the input of Rx (RCVR)=> $N=kT_0 B$
- B: Bandwith (Hz)
- T=Temperature (standard condition: $T_0 = 290\,^oK = 17^oC$)
- k=Boltzman's constant ($1.30\ 10^{-23}\ J/^oK$)

Thermal noise power (B=1 Hz): $N=kT_0 \cong$ -204 dB (noise power level)

S_{min} and SNR



Noise power at the input of the Rx: $N_{\text{in}} = kT_0 B$

Input signal to the Rx: $S_{\text{in}} = \dfrac{kT_0 B F_n S_{out}}{N_{out}}$ $\qquad F_n = \dfrac{S_{\text{in}}/N_{\text{in}}}{S_{out}/N_{out}}$

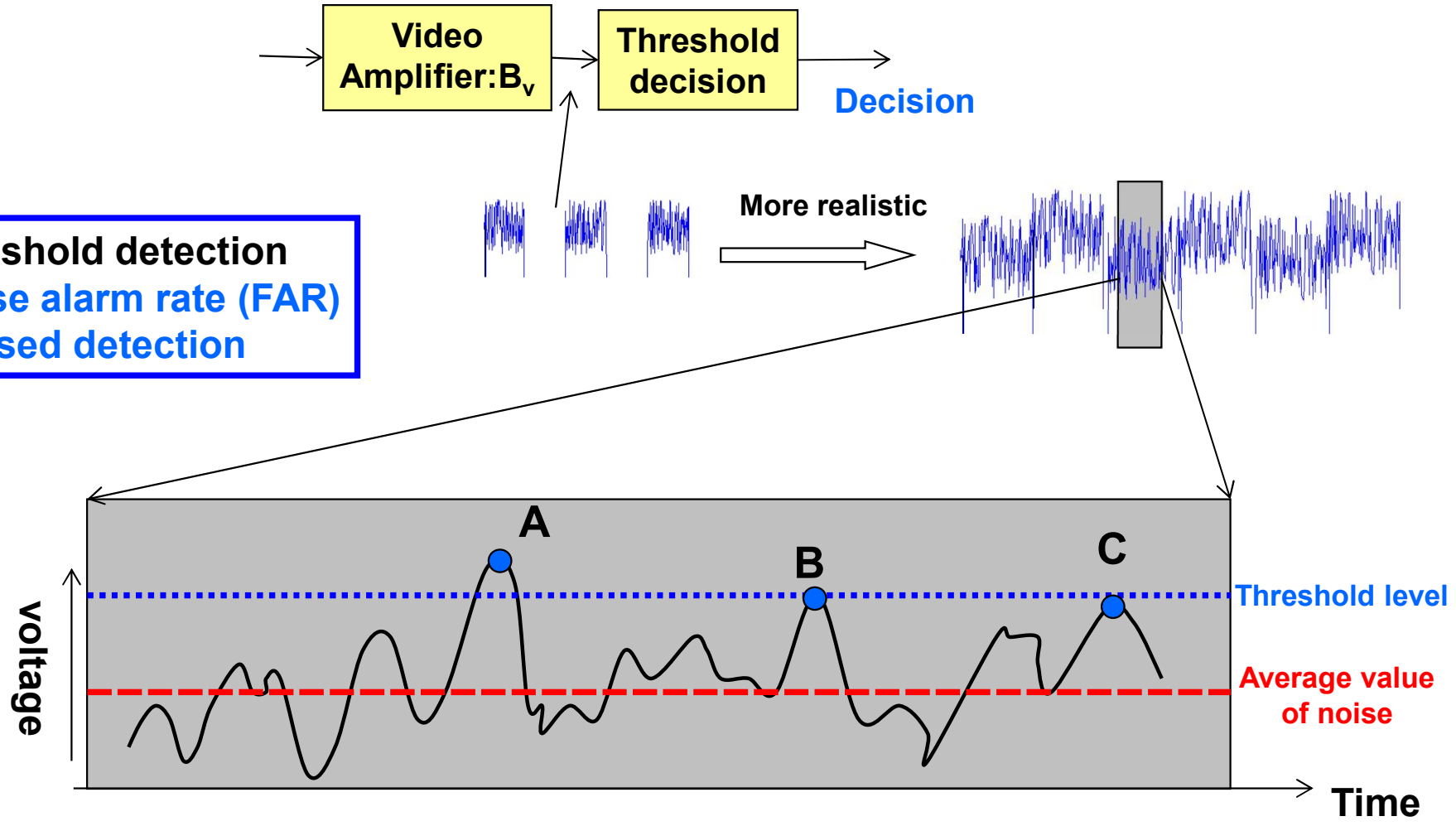
Minimum detectible SNR at the output of the Rx: $(S_{out}/N_{out})_{\min}$

$$(S_{out}/N_{out})_{\min} = (S_{out}/N_{out}) \text{ when } S_{\text{in}} = S_{\min}$$

giving

$$S_{\min} = kT_0 B F_n \left(\frac{S_{out}}{N_{out}}\right)_{\min}$$

Threshold detection and False Alarms



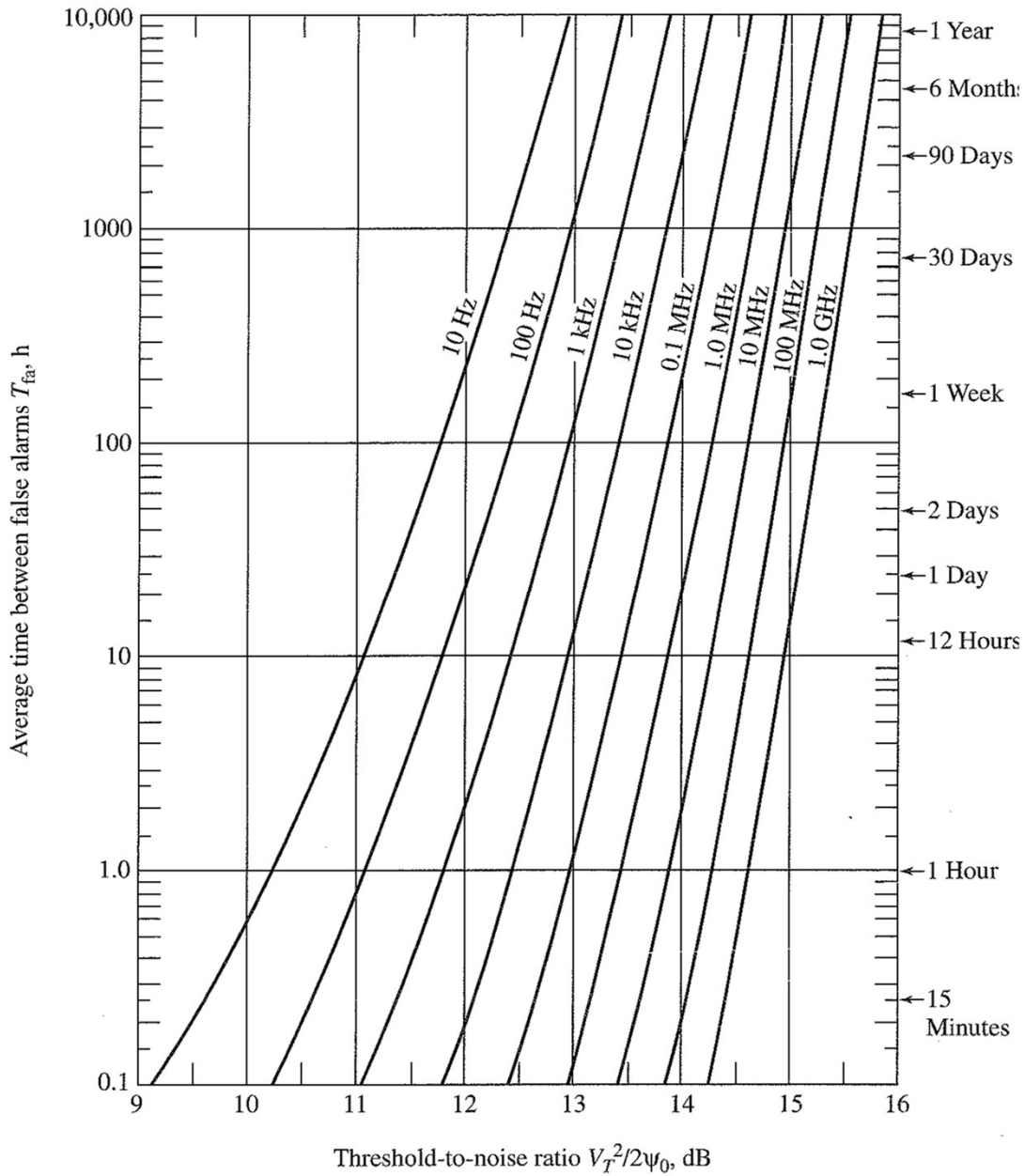
P_{fa}=probability of false alarms

Probability of False-alarm [Skolnik]

- Gaussian noise without signal at the input to the detector
- Probability that noise level exceeds the threshold: propability of false alarm
- The time between the false alarms is a good measure of the effect of noise on radar perfomance

Example:
In 1MHz BW = 1μs PW
1 million decisions/s
1 false alarm per second on average
=>false alarm probability of 10^{-6}

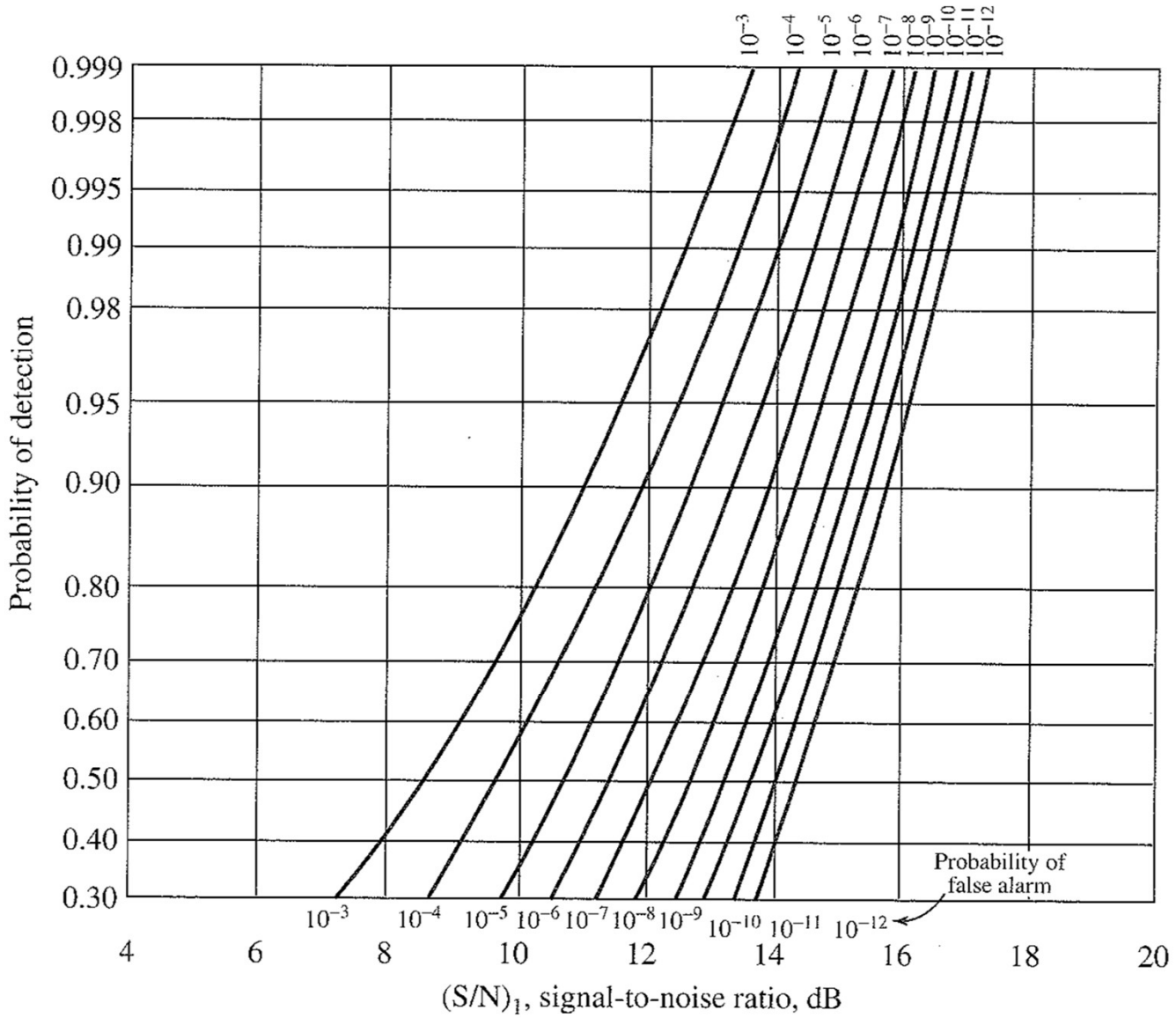
ψ_0: mean power of noise
V_T: threshold voltage



Probability of detection [Skolnik]

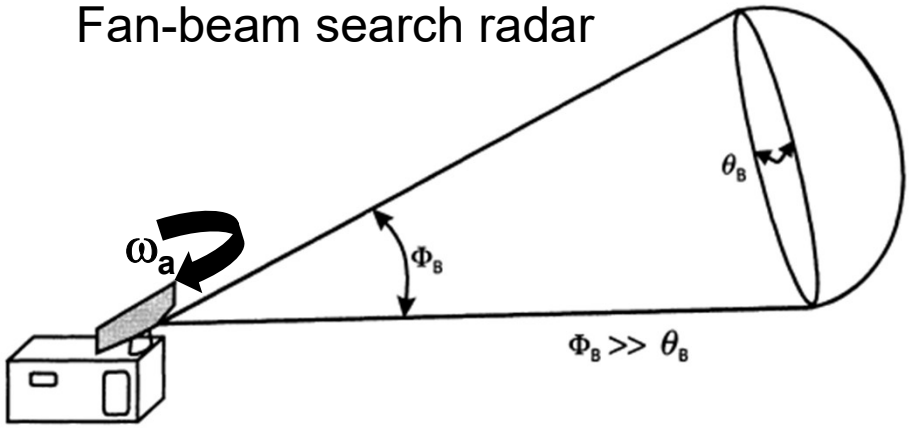
Detection of sin wave with noise

3dB difference in SNR
0.99=>0.50



Pulse integration

Fan-beam search radar



- Peak power of pulse: P_t
- Pulse duration: τ
- PRF: f_p
- Time on target(dwell time): $T_{ot} = \theta_B/\omega_a$
- Single echo pulse energy : $E_s = P_t \tau$
- Pulse per scan (hits per scan): n

$$n = T_{ot}\, f_p = \frac{\theta_B f_p}{6\omega_a} \quad \text{(integer)}$$

$\omega_a \equiv$ revolutions per minute (rpm)
(5 rpm≡5x**6**(=360/60)=30^0/s)

- Radar has more than one echo pulse during T_{ot}
- When echo pulses are integrated, the energy received $E_n = nE_s$
- This increases probability of detection, and reduces the false alarm rate.
- So, use it!

Example: air surveillance radar

A long range ground-based air surveillance radar

- 340 Hz PRF
- 1.5^0 beamwidth
- 5rpm antenna rotation rate
- PRI=?
- Unambiguous range R_{un}=?

Consider pulse integration for this radar

- Tot=?
- Number of pulses that can be integrated n=?

The radar range equation

$$R_{max}^4 = \frac{P_t G A_e \sigma}{(4\pi)^2 k T_0 B F_n (S/N)_{min}} \Rightarrow R_{max}^4 = \frac{P_t G A_e \sigma}{(4\pi)^2 k T_0 B F_n (S/N)_n}$$

Integration efficiency: $E_i(n) = \frac{(S/N)_1}{n(S/N)_n}$ $\Rightarrow$ $R_{max}^4 = \frac{P_t G A_e \sigma n E_i(n)}{(4\pi)^2 k T_0 B F_n (S/N)_1}$

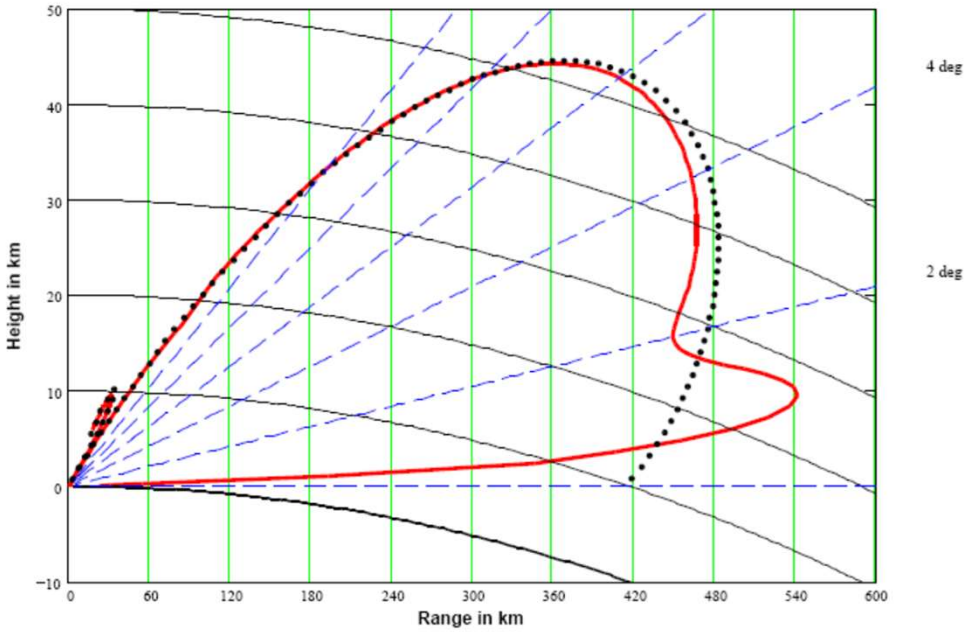
Additonal Losses

- Transmission losses: Tx to antenna, and between the elements
- Receiver losses: line and elements between the receiver and antenna
- Beamshape losses: gain is not constant during the time on target (dwell time)
- Filter losses (matched filter): losses introduced by imperfect filter
- Signal processing losses
- RCS Fluctuation losses (introduced as Swerling target model)

All these losses must be taken into account, and would reduce the range of radar!

Example: Early Warning Radar System

Tall King Coverage over Farmland, h_a = 20m



VHF EW Radar: Tall King

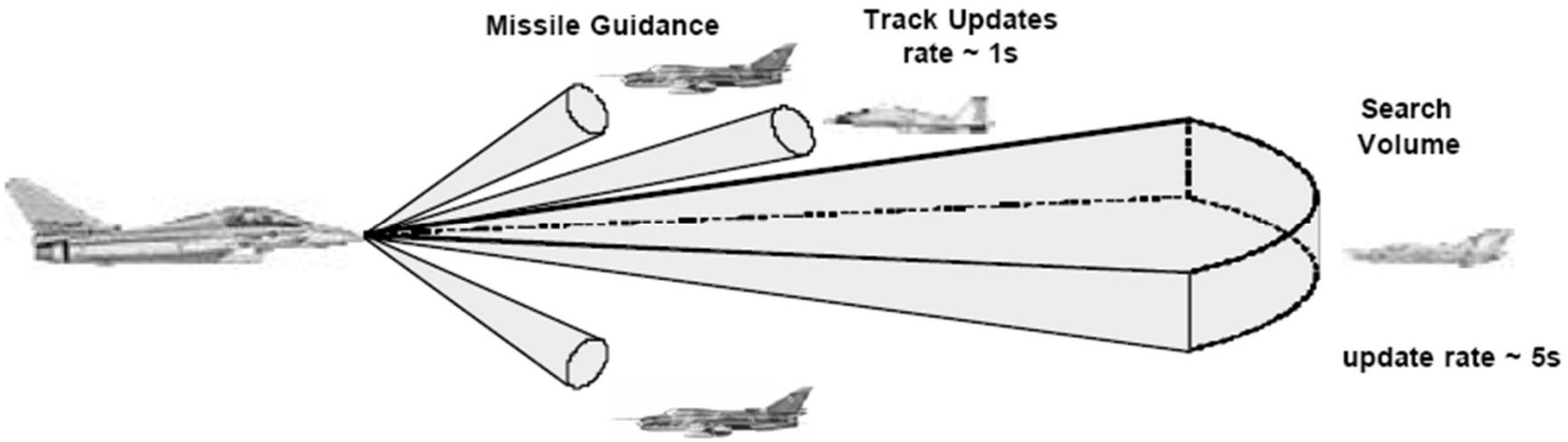


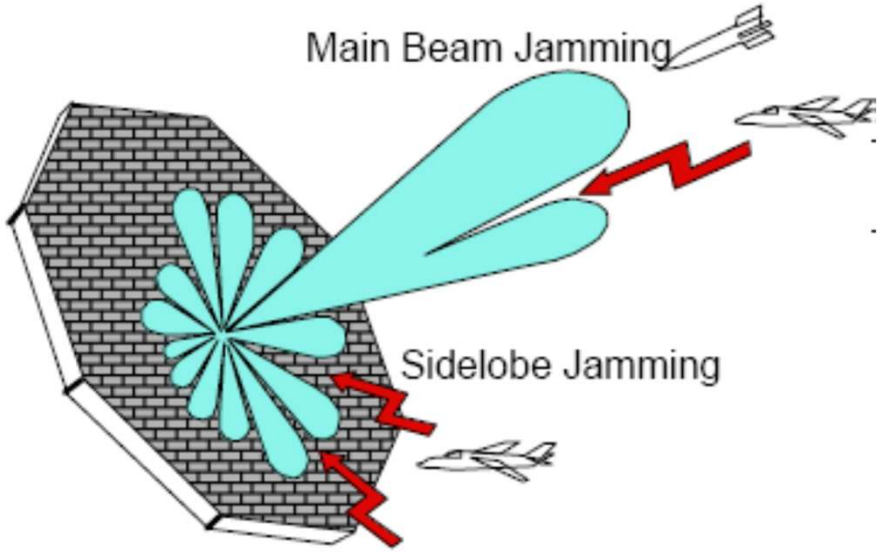
VHF band $\lambda = 1.8\text{m}$ $P_{av} = 10\ \text{kW}$

$w = 32\text{m}$ $h = 12.5\text{m}$ $A_r = 240\ \text{m}^2$ $P_{av}A_r = 2400\ \text{W-m}^2$ $R_m = 460\ \text{km}$

$\theta_a = 3.5°$ $\theta_e = 8.9°$ $t_s = 10\text{s}$

Example: Airborne radar tracking





Parameter	Value
Peak RF power	8 kW
Mean RF power	1 kW
Noise Figure	5 dB
Array Face Size (w x h)	550 mm x 550 mm
Volume Envelope (w x h x d)	600 x 600 x 200 mm
Weight	75 kg
Power Consumption	7 kW
Heat Dissipation	6 kW

Example: Air Traffic Control Radar



Type:	Enroute 2D air surveillance radar
Band:	L
Frequency:	1250 to 1350MHz
Peak Power:	5MW
Antenna size:	12.8x6.7m
Antenna Gain:	36dB (lower beam) 34.5dB (upper beam)
Beam shape:	$Cosec^2$
Elev beamwidth:	4° $Cosec^2$ to 40°
Azim beamwidth:	1.25°
Scan:	Mechanical
Scan Rate:	6rpm
PRF:	360pps
PRF Stagger:	Quadruple
Pulse width:	2μs
Noise Figure:	4dB

Example: A typical civil marine radar

- Frequency=9400 MHz (X band)
- Antenna
  - Horizontal (azimuth) beamwidth=0.8^0
  - Vertical(elevation) beamwidth=15^0
  - Gain=33dB
- Azimuth rotation rate=20rpm
- Peak power=25 kW
- Pulse width=0.15μs
- PRF=4000Hz
- Receiver Noise Figure=5dB
- Receiver BW=15MHz
- System Losses=12dB

AN Nomenclature (Joint Army-Navy Nomenclature System)

Firts Letter:Installation
A - Piloted Aircraft
B - Underwater Mobile(submarine)
C - Cryptographic Equipment (NSA)
D - Pilotless Carrier (drone, UAV)
F - Fixed Ground
G - General Ground Use
K - Amphibious
M - Ground Mobile
P - Human Portable
S - Water (surface ship)
T - Transportable (ground)
U - General Utility (multi use)
V - Vehicle (ground)
W - Water Surface+Underwater
Z - Piloted/Pilotless +Airborne

Second letter: type of equipment
A - Invisible Light, Heat Radiation
B - Comsec (NSA use only)
C - Carrier (electronic wave or signal)
D - Radiac (Radioactivity Detection)
E - Laser
F - Fiber Optics (was Photographic)
G - Telegraph or Teletype
I - Interphone and Public Address
J - Electromechanical or inertial wire
K - Telemetering
L - Countermeasures
M - Meteorological
N - Sound in Air
P - Radar
Q - Sonar and Underwater Sound
R - Radio
S - Special or Combination
T - Telephone (Wire)
V - Visual, Visible Light
W - Armament (not otherwise covered)
X - Fax or Television
Y - Data Processing
Z - Communications (NSA use only)

Third letter: purpose
A - Auxiliary Assembly
B - Bombing
C - Communications (two way)
D - Direction Finding, Reconnaissance and Surveillance
E - Ejection and/or Release
G - Fire Control or Searchlight Directing
H - Recording and/or Reproducing
K - Computing
L - no longer used. Was Searchlight Control, now covered by "G".
M - Maintenance or Test
N - Navigation Aid
P - no longer used. Was Reproducing, now covered by "H"
Q - Special or Combination
R - Receiving or Passive Detecting
S - Detecting, Range and Bearing, Search
T - Transmitting
W - Automatic Flight or Remote Control
X - Identification or Recognition
Y - Surveillance (target detecting and tracking) and Control (fire control and/or air control)
Z - Secure (NSA use only)

AN/APG-63: Airborne Radar, Fire control, model #63
AN/SPS-55: Ship borne Search radar, model #55
AN/BQR-2 : Underwater Sonar, for search or ranging, model #2